Fluid–Structure Interaction of 3D-Printed Flexible Underwater Vehicle Fins

Dual-Fidelity Modeling, Deformed-Camber OpenFOAM/CalculiX Verification, and Polymer Design Study

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1. Motivation & Core Physical Challenges

Why 3D-Printed Flexible Underwater Fins?

- Bio-inspired AUVs use compliant polymers (**TPU 85A/95A, PA12 Nylon, PETG, PLA**).
- Passive bending toward the flow (**hydroelastic washout**) sheds excess lift, cuts drag, and reduces root stress by $> 60\%$.

Three Engineering Challenges

1. **Added-Mass Instability ($\mu \ll 1$)**: $\rho_s \approx \rho_f \Rightarrow \mu = \frac{\rho_s h_{\text{mean}}}{\rho_f L} \approx 0.066$. Water added mass is $12\times$ fin mass $\Rightarrow$ explicit coupling diverges in 1–2 steps.
2. **3D-Print Anisotropy & Moisture**: Shell–infill sandwich mechanics + PA12 Nylon loses 55% stiffness in water.
3. **Speed vs. Fidelity**: Need < 1 s interactive runs for design sweeps *and* full Navier–Stokes + 3D FEA verification.

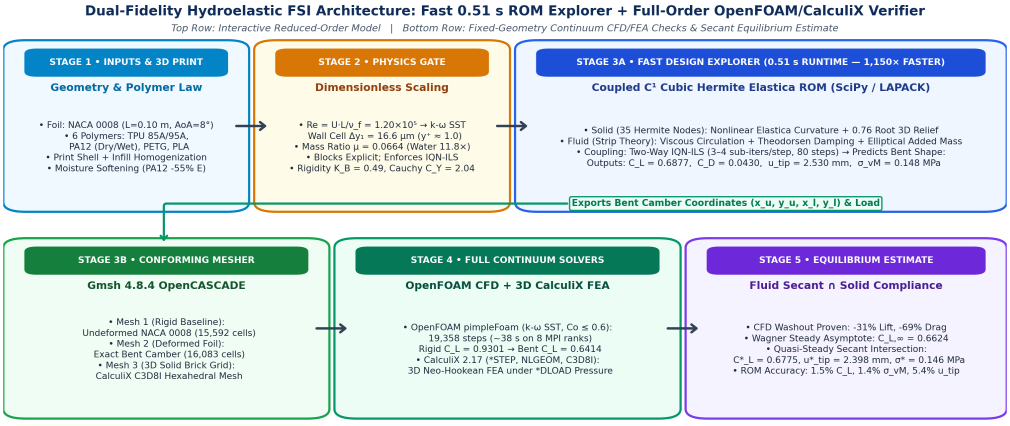
Governing Dimensionless Numbers

Group	Definition & Role
$Re = \frac{U_\infty L}{\nu_f}$	Laminar (< 2k) vs. k–ω SST
$\mu = \frac{\rho_s h_{\text{mean}}}{\rho_f L}$	Added-mass ratio (≈ 0.03–0.15)
$K_B = \frac{D_{\text{mean}}}{\rho_f U_\infty^2 L_{\text{bend}}^3}$	Dimensionless bending stiffness
$C_Y = 1/K_B$	Cauchy number (washout driver)
$U_r = \frac{U_\infty}{f_{1,\text{water}} L_{\text{bend}}}$	Reduced velocity (flutter gate)

Submerged Frequency Drop

$$f_{1,\text{water}} = \frac{f_{1,\text{vacuum}}}{\sqrt{1 + \Gamma_{3D} \frac{\pi}{4\mu}}} \approx 0.28\, f_{1,\text{vacuum}}$$

2. Two Solver Paths for the Same Hydroelastic Problem



Full-order path: Gmsh → OpenFOAM k–ω SST → CalculiX 3D `NLGEOM IQN-ILS` (0.51 s)

Reduced-order path: C^1 Hermite beam +

3. What Each Solver Computes

Full-Order Solver: OpenFOAM + CalculiX

- **OpenFOAM**: unsteady k–ω SST CFD on prescribed rigid and bent meshes; returns C_L, C_D, C_M.
- **CalculiX**: 3D `C3D8I NLGEOM` FEA under hydrodynamic load; returns u_{tip} and σ_{vM}.
- Supplies the physical reference: deformed-camber loading, 3D stress, and the 15-case material campaign.

Equilibrium Estimate

The decreasing OpenFOAM fluid line and increasing CalculiX compliance line are intersected. This is a quasi-steady secant estimate; a fully time-accurate moving-mesh run remains open work on the full-order solver.

Coupled Reduced-Order Solver (0.51 s)

- Solves fluid load and 1D beam bending together.
- Uses 35 C^1 Hermite nodes, 80 time steps, and `IQN-ILS`.
- Carries the same physics: added mass, washout, elastica curvature, hyperelastic-relief calibration.
- Direct outputs: bent shape, C_L, C_D, C_M, u_{tip}, and σ_{vM}.

Why `IQN-ILS`?

Water added mass is 11.8$\times$ the fin mass, so explicit exchange diverges; `IQN-ILS` converges in 3–4 sub-iterations.

4. Full-Order Evidence Chain: What Each Column Means

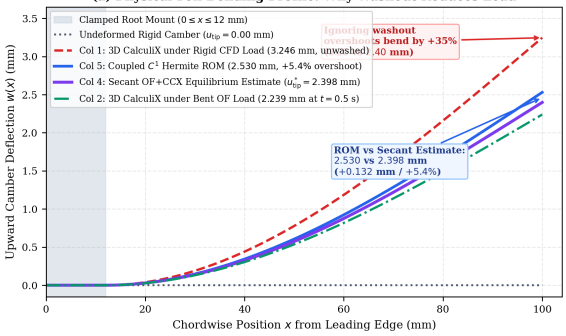
Benchmark Case: TPU 95A NACA 0008, $L = 0.10$ m, $U_\infty = 1.2$ m/s, $\alpha_0 = 8°$, $Re = 1.20 \times 10^5$, $\mu = 0.0664$ (60-vCPU GCP Spot VM)

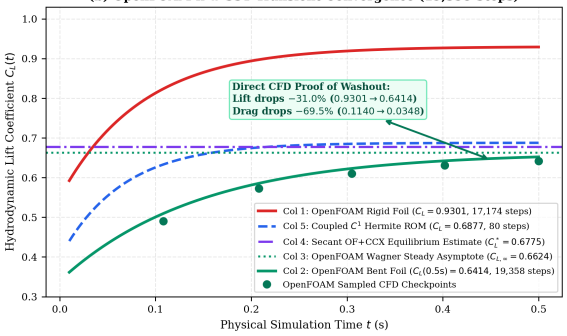
Evidence Stage	**What Is Actually Computed**	C_L	$u_{\mathbf{tip}}$	$\sigma_{\mathbf{vM}}$
1. Rigid baseline	OpenFOAM on unbent mesh; load passed to CalculiX	0.9301	3.246 mm	0.197 MPa
2. Bent check (0.5 s)	OpenFOAM on reduced-order bent mesh; load passed to CalculiX	0.6414	2.239 mm	0.136 MPa
3. Steady bent estimate	Wagner extrapolation of Stage 2	0.6624	2.317 mm	0.141 MPa
4. Secant equilibrium	Fluid secant $\cap$ solid compliance; not a separate run	**0.6775**	**2.398** mm	**0.146** MPa
5. Coupled ROM	Direct reduced-order output (35 nodes, 80 steps, 0.51 s)	**0.6877**	**2.530** mm	**0.148** MPa

Key Takeaway from the Full-Order Solver

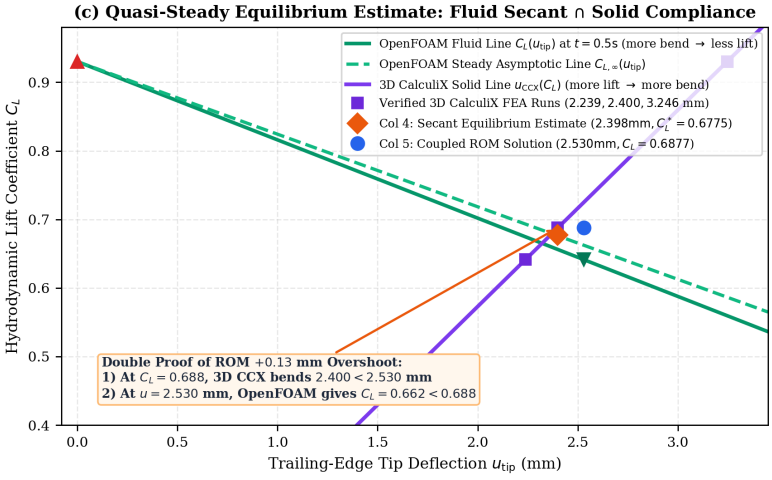
- Re-meshing the bent camber in Gmsh and solving 19,358 steps of OpenFOAM `pimpleFoam` establishes that passive washout drops lift by $-$**31%** (0.9301 $\rightarrow$ 0.6414–0.6624) and drag by $-$**69%**.
- Against the secant estimate built from those states, the reduced-order model differs by **1.5%** in C_L, **1.4%** in σ_{vM}, and **5.4%** in u_{tip}.

5. Full-Order Evidence: Bending Changes the Load

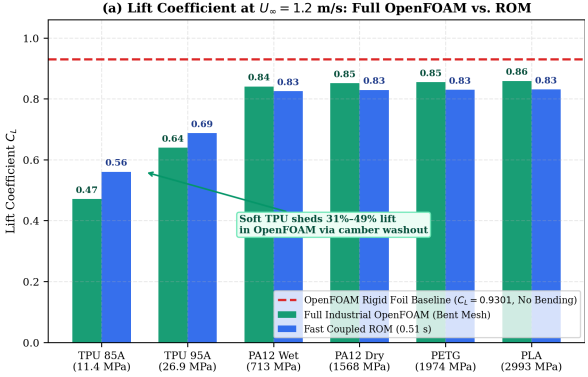


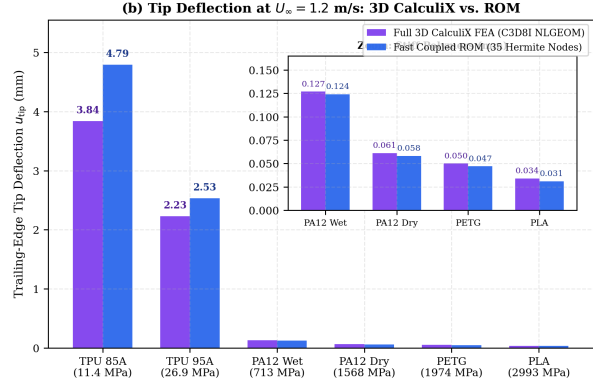


6. From Computed States to a Quasi-Steady Equilibrium Estimate

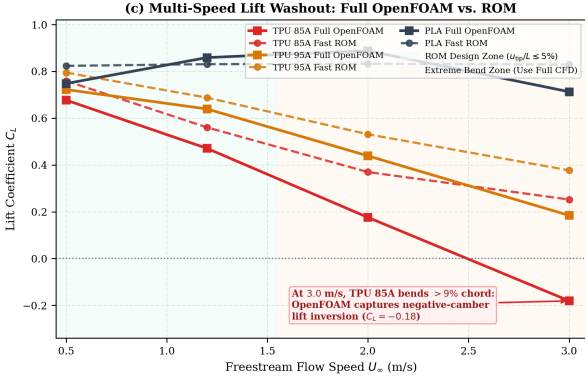


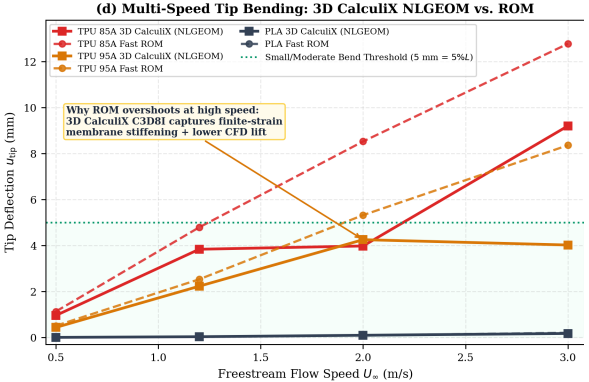
7. Full-Order Campaign at the Design Speed: Six Polymers



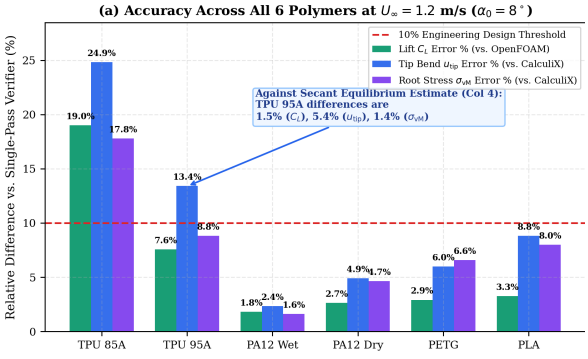


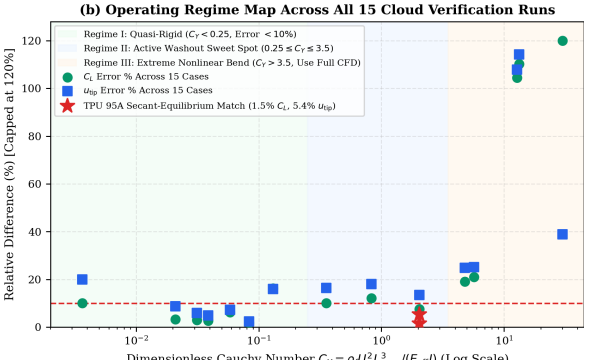
8. Beyond the Reduced-Order Range: Extreme TPU Bending



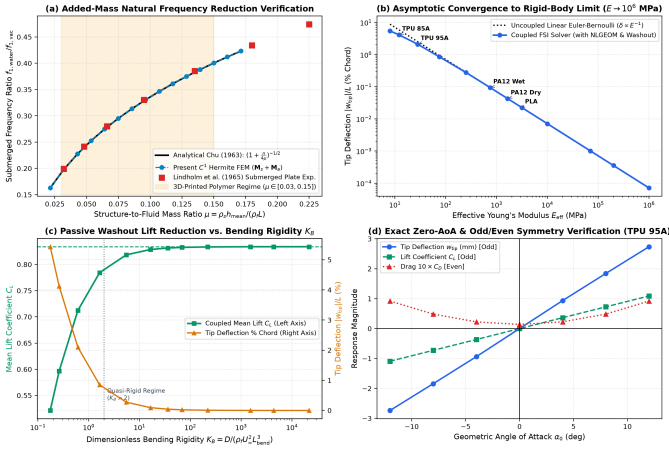


9. Quantified Accuracy and Dimensionless Operating Envelope



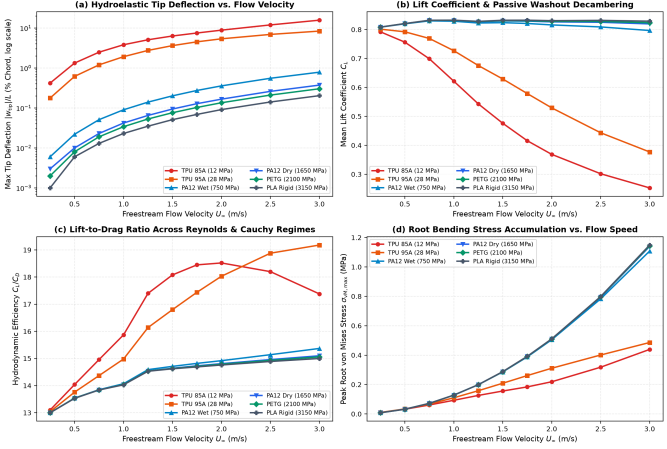


10. Analytical Benchmarks for the Reduced-Order Solver



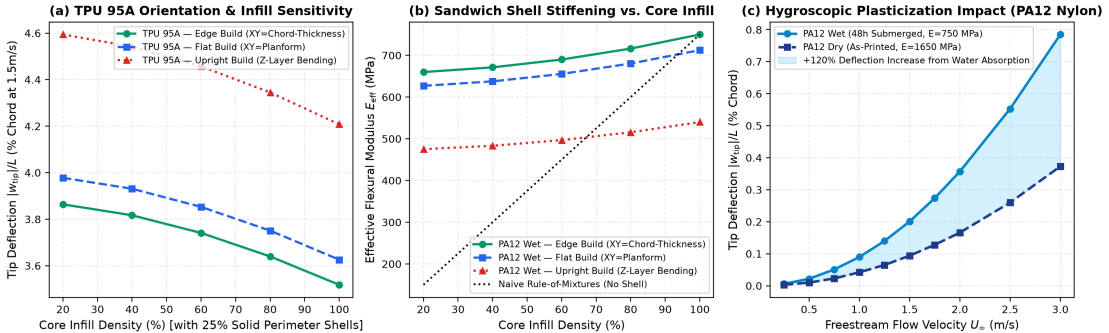
11. Design Study: Six Polymers Across Flow Velocity

Figure 2: Multi-Polymer Hydroelastic Performance Trends Across Flow Speed ($L = 0.10\text{m}, t/c = 0.08, \alpha_0 = 8^\circ$)



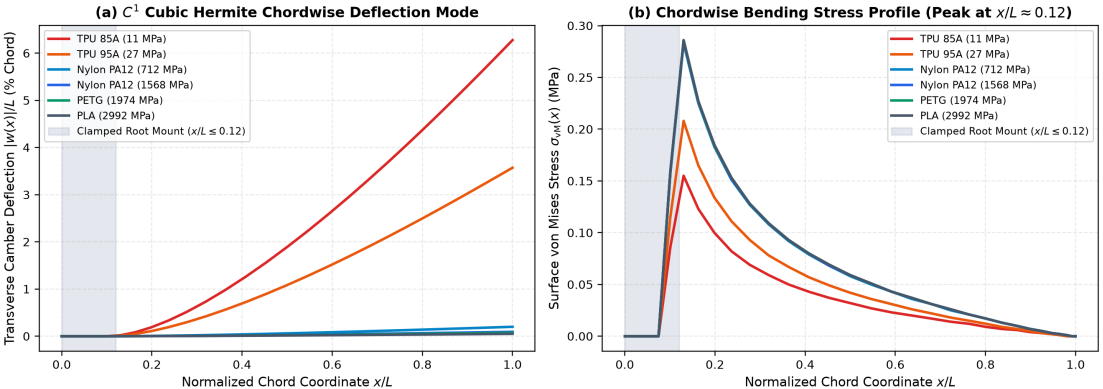
12. Manufacturing Insight: Orientation, Infill, and Wet Nylon

Figure 3: Additive Manufacturing Anisotropy (Build Orientation, Shell-Infill Homogenization & PA12 Moisture Plasticization)



13. Where Does the Fin Bend and Where Is Stress Highest?

Figure 4: Chordwise Deformed Camber Line & Surface Bending Stress Distribution ($U_\infty = 1.5$m/s, $\alpha_0 = 8^\circ$)



14. Conclusions, Limitations, & Future Scope

Summary of Achievements

1. **Dimensionless Governance**: Automated Re, μ, K_B, C_Y, U_r regime classification and $y^+ = 1$ meshing, applied identically to both solver paths.
2. **Full-Order Continuum Solver**: Deformed-camber Gmsh–OpenFOAM–CalculiX chain establishes a -31% passive-washout lift drop and a 15-case polymer/speed reference envelope.
3. **Coupled Reduced-Order Solver**: Multi-secant `IQN-ILS` remains stable in 3–4 sub-iterations at $\mu = 0.066$ and reproduces the full-order response within 1.4–5.4% for $C_Y \leq 3$.
4. **Interactive Web Platform**: Real-time 2D/3D visualizers, ParaView VTK export, and full solver deck generation.

Future Work (Next Steps)

1. **Complete the Full-Order Solver**: Continuous two-way preCICE coupling with `dynamicMotionSolverFvMesh`, replacing the secant estimate with a running moving-mesh solve.
2. **Water-Tunnel Experiments**: PIV wake measurement and waterproof root strain-gauge testing on 3D-printed TPU 95A and PA12 fins, bounding both solver paths.
3. **Cavitation & Viscoelasticity**: Schnerr–Sauer multiphase cavitation ($U_\infty > 6$ m/s) and Prony-series polymer creep/fatigue.
4. **Graded Infill Optimization**: Spatially varying infill $\phi(x, z)$ to maximize C_L/C_D.